32. The Mathematical Tree

The Hyper-operation Sequence

Successor is a primitive function: it is defined in terms of no simpler function.

$$x + 1$$

Addition is defined recursively in terms of successor.

$$x + y = x + \underbrace{1 \ldots + 1}_{y \text{ times}}$$

y copies of 1 added to x

Multiplication is defined recursively in terms of addition.

$$x \times y = \underbrace{x + x \ldots + x}_{y \text{ times}}$$

y copies of x combined by addition

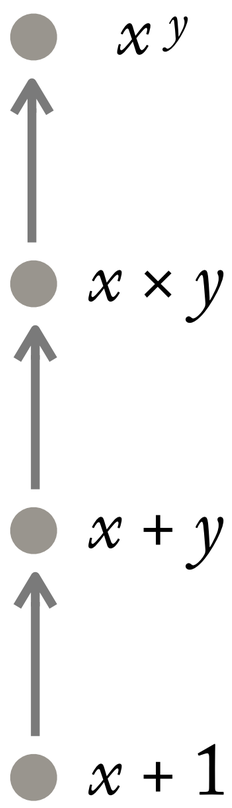
Exponentiation is defined recursively in terms of multiplication.

$$x^y = \underbrace{x \times x \ldots \times x}_{y \text{ times}}$$

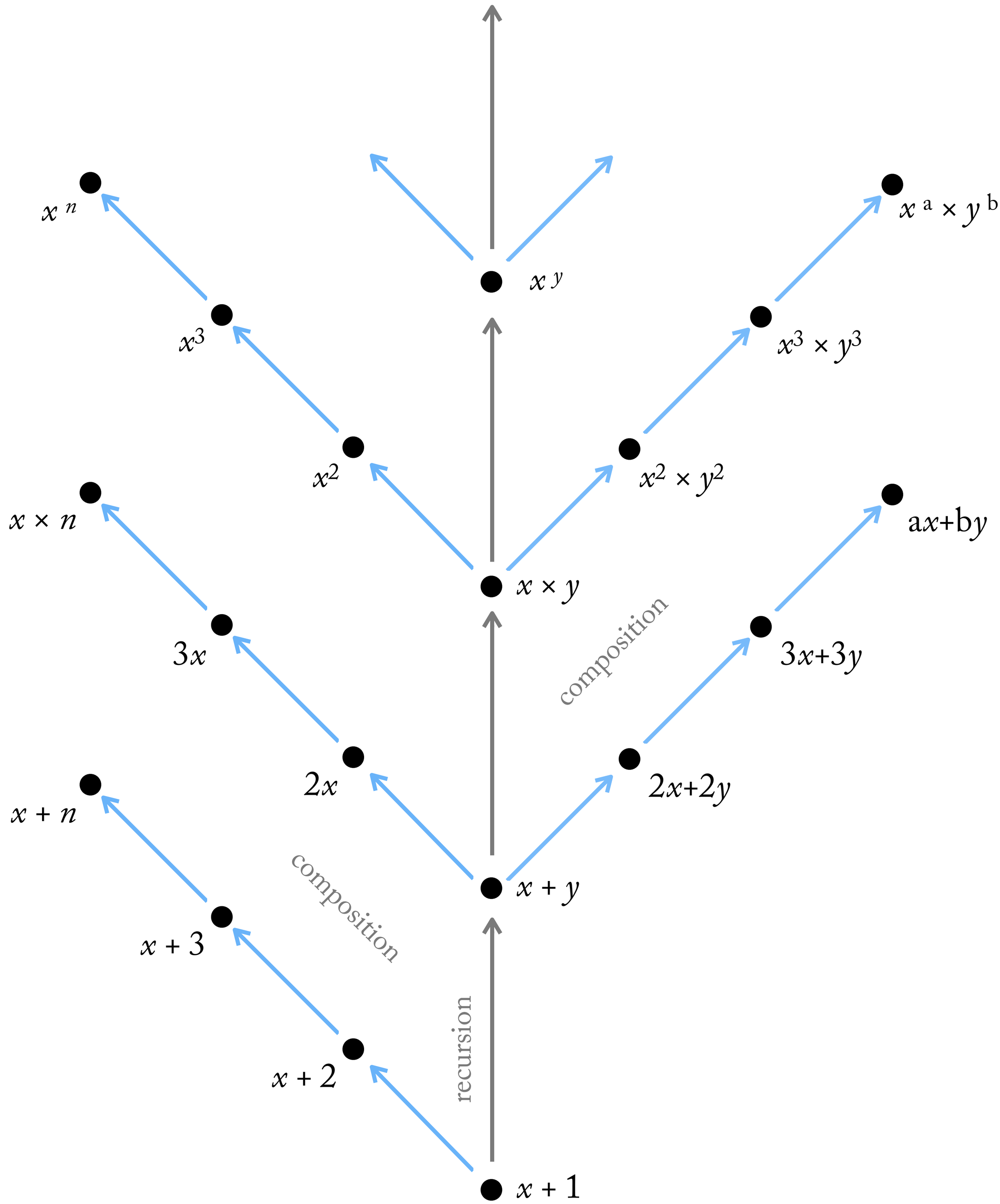
y copies of x combined by multiplication

Through recursion, we can build steadily up from successor to the further reaches of mathematical space.

Each function in the sequence can be computed by an algorithm that relies on a sub-routine for the previous function in the sequence.

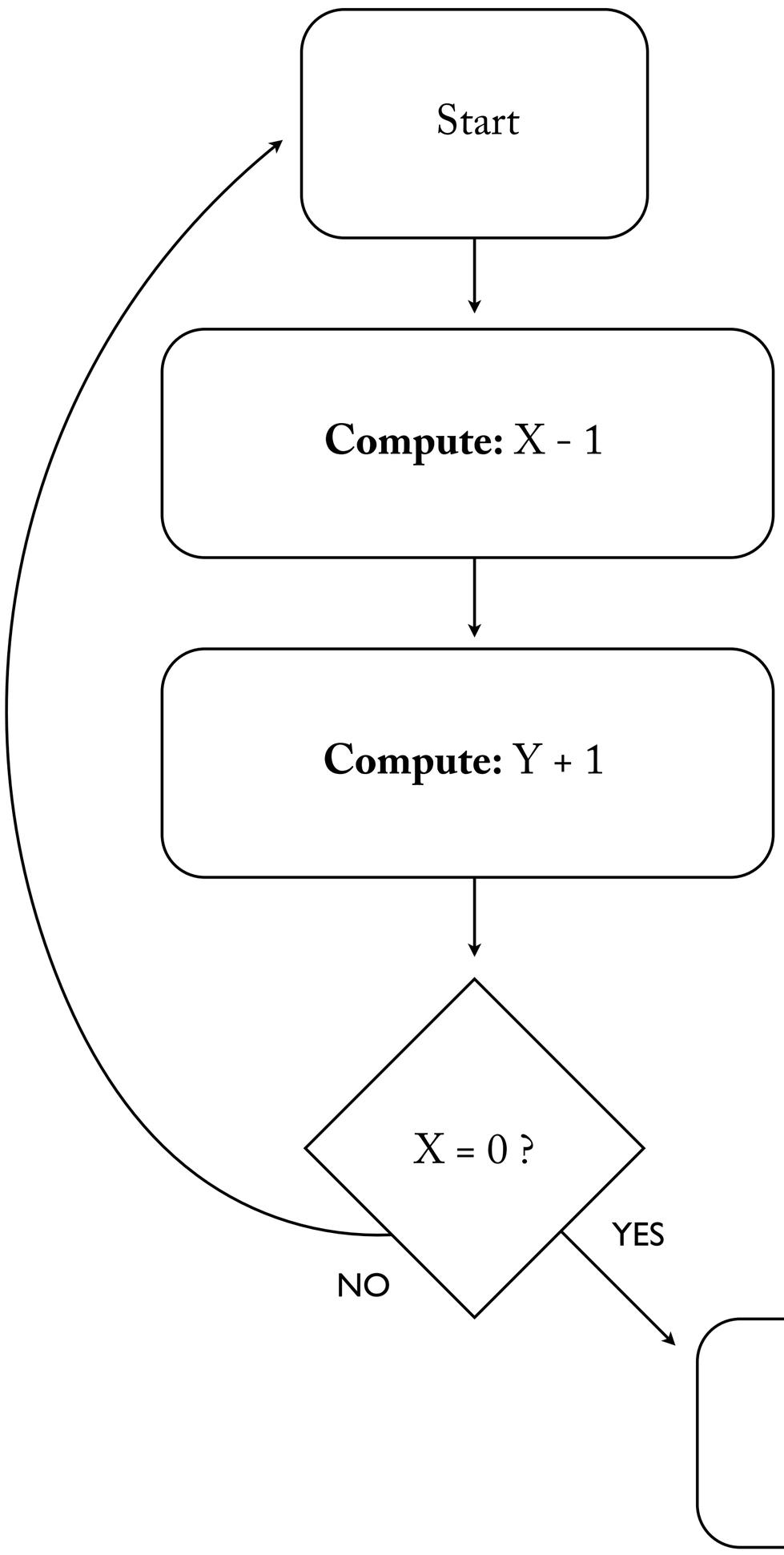


Recursion + Composition = Computable Mathematics



Recursive Addition

Compute $f(x,y) = x + y$



How can we break addition up into a simple, repeating operation?

Adding x to y is equivalent to adding 1 to x, y times.

$$
\begin{aligned}
2 + 3 &= 2 + 1 \\
&= 3 + 1 \\
&= 4 + 1 \\
&= 5
\end{aligned}
$$

3 times

$$
\begin{aligned}
4 + 5 &= 4 + 1 \\
&= 5 + 1 \\
&= 6 + 1 \\
&= 7 + 1 \\
&= 8 + 1 \\
&= 9
\end{aligned}
$$

5 times

